

Credit Card Fraud Detection using Hidden Naive Bayes and Bayesian Belief Network

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Abstract—This paper evaluates credit card fraud detection models using Hidden Naive Bayes (HNB) and Bayesian Belief Network (BBN) approaches. It aims to design accurate models for identifying fraudulent transactions in a dataset with various transaction features. HNB and BBN use probabilistic reasoning and statistical analysis to classify transactions as fraudulent or not based on these features. Performance metrics are employed to evaluate the models. Results show HNB achieving 86.87% accuracy, and BBN reaching 89.59%. Additionally, a comparison with other fraud detection approaches highlights HNB and BBN's competitive performance. Overall, this paper showcases their potential in accurately detecting credit card fraud, offering valuable tools for this endeavor.

Keywords—Credit Card fraud, classification, Prediction, Hidden Naive Bayes, Bayesian Belief Network, probabilistic reasoning, statistical analysis

I. INTRODUCTION

In the contemporary digital landscape, credit cards have emerged as indispensable tools for conducting financial transactions, offering unparalleled convenience, flexibility, and global reach. These small plastic cards have revolutionized the way we engage in commerce, from everyday purchases to high-value transactions. However, this widespread reliance on credit cards has given rise to a persistent and ever-evolving menace: credit card fraud.

Credit card fraud poses a substantial threat, not only to individual consumers but also to the stability and security of financial institutions and the broader economy. Cybercriminals, armed with increasingly sophisticated tactics, continually seek vulnerabilities within digital payment systems to perpetrate fraudulent transactions. The consequences are profound, ranging from financial losses and data breaches to the erosion of trust in electronic payment systems.

It is in this context that the development and implementation of advanced fraud detection systems become imperative. These systems are not a mere convenience but a critical necessity in the contemporary financial landscape. They serve as the first

line of defense against the evolving tactics of cybercriminals, mitigating financial losses, preserving data security, and upholding the trust that underpins the global financial ecosystem.

This paper is dedicated to addressing this pressing need through the development and evaluation of credit card fraud detection models, specifically leveraging the Hidden Naive Bayes (HNB) and Bayesian Belief Network (BBN) methodologies. These models are chosen for their adeptness at probabilistic reasoning and statistical analysis, making them well-suited for unraveling the intricate patterns concealed within vast repositories of transactional data.

The objective of this paper is dual-fold: first, to engineer models that excel in the identification of fraudulent credit card transactions with accuracy, and second, to distinguish these illicit activities from legitimate transactions. Striking this balance is paramount, as false positives can inconvenience customers and erode trust, while false negatives can lead to substantial financial losses.

In this pursuit, we delve into the application of HNB and BBN models, exploring their potential to serve as formidable tools in the battle against credit card fraud. By enhancing financial security, preserving trust, offering effective countermeasures to combat evolving threats, and bolstering the defenses of the digital financial realm, this paper endeavors to safeguard the interests of consumers, businesses, and the broader economy.

II. RELATED WORK

The research methodology initiated with an extensive study of existing literature focused on employing ML techniques to detect credit card fraud. Multiple scholarly papers were analyzed to gain insights into how Machine Learning Models and Artificial Neural Networks (ANN) algorithms are utilized for fraud detection in credit card transactions. This comprehensive literature review provided a deeper understanding of various approaches, algorithms, and feature sets used in predicting

credit card fraud, guiding subsequent stages of the study, especially in selecting algorithms and methodologies.

An efficient fraud detection system introduced by Trivedi et al. [1] using Gradient Boosting and Random Forest showcased prominent results. Both the models were evaluated with the help of European cardholders dataset. The study focused mainly on accuracy and precision where GB achieved 94.01% accuracy and a precision of 93.99%, while RF attained 94.00% accuracy and an impressive 95.98% precision.

ML based framework for detecting credit card frauds was proposed in a paper by Tanouz et al. [2]. Their study utilized the European cardholders dataset and addressed the class imbalance issue through under-sampling techniques. Random Forest model showcased an accuracy of 91.24% in detecting frauds. In comparison, Logistic Regression model performed very well with an accuracy of 95.16%. Additionally, the authors emphasized the need for further investigation into class imbalance challenges using the confusion matrix.

Neural network algorithms like Extreme Learning Machine (ELM) and Multilayer Perceptron (MLP) were employed by Riffi et al. [3] to derive better fraud detection systems. The European cardholders dataset from 2013 was used in training and evaluating these models. These ANNs differ largely in their internal structure but worked very well in finding fraud transactions with an accuracy of 95.46% for ELM and even better accuracy of 97.84% for MLP. The study concluded that while MLP demonstrated superior performance, ELM offered a simpler architecture.

Ensembling Learning algorithms, Adaptive Boosting (AdaBoost) and Majority Voting (MV) were utilized by Randhawa et al. [4] in developing a standard fraud detection system. These ensembling methods were applied on ML models such as Support Vector Machine improve the fraud detection capacity of the models. The models were trained and tested against the European cardholders dataset. The AdaBoost-SVM approach showcased an accuracy of 99.959% with an MCC value of 0.044.

A Comparative study including ML methods such as Random Forest and K-Nearest Neighbour was conducted by Rajora et al. [5] to derive insights on the working efficiency of different models for the purpose of detecting credit card frauds. The same European cardholders dataset was used here as well. Incidentally, these models attained better accuracy and Area Under the Curve (AUC) values which were the primary metrics considered. RF outperformed KNN with an accuracy of 94.9% and AUC of 0.94, whereas the latter attained the an accuracy of 93.2% and an AUC of 0.93. The class imbalance problem present in the dataset was thoroughly addressed in this research.

Similar to Rajora et al. [5], the research by Khatri et al. [6] also included investigation of various commonly used ML models in the scope of detecting credit card frauds. Their research showcased the ability of each model in classifying the a transaction as fraud or non-fraud. The assessment utilized a dataset generated from European cardholders in 2013. Though five different models were researched (DT, KNN, RF, LR,

Naive Bayes), the one with the outstanding performance was KNN. It stood out from other models as it was able to precisely flag the fraud transactions as fraud. It achieved a precision of 91.11% and a sensitivity of 81.19%.

III. PROPOSED METHODOLOGY

The methodology for credit card fraud detection using Hidden Naive Bayes (HNB) and Bayesian Belief Network (BBN) algorithms involves several key steps. Firstly, the credit card transaction data is preprocessed, including cleaning, normalization, and feature engineering, to prepare it for modeling. Then the dataset is split into training and testing set. Since the dataset is highly imbalanced Random Over Sampling (ROS) is performed on the training set to balance the training data before training of the model. Next, a Hidden Naive Bayes model is trained on the preprocessed data, incorporating hidden variables to capture dependencies between observed features. This model is able to estimate the likelihood of fraud for each transaction. Simultaneously, a Bayesian Belief Network is constructed using the preprocessed dataset, defining the network structure and estimating conditional probability distributions for each node based on the training data. The BBN captures the complex relationships and dependencies between different features. The predictions from the HNB and BBN models are used to classify the new data. The two models were evaluated using accuracy, precision, recall, and F1-score. Fine-tuning and optimization techniques are applied to optimize the individual HNB and BBN models. This involves exploring parameter tuning methods to enhance their performance. It is integrated with existing fraud detection systems and workflows to ensure smooth adoption. The model is continuously monitored and updated to adapt to evolving fraud patterns and maintain its effectiveness over time. Overall, this methodology combines the power of HNB, and BBN algorithms to create a robust system for credit card fraud detection that accurately flags fraudulent transactions while minimizing false positives and handling complex patterns in a real-time and scalable manner.

A. Dataset

"Abstract dataset for Credit card fraud detection" [7] provided by Larsen & Toubro Infotech was used for the proposed method. The dataset has 11 features and 3075 instances.

B. Model Implementation

- **Load Dataset:** Acquire and load the credit card transaction dataset, serving as the foundation for fraud detection. It contains transaction details like amounts, timestamps, and location data.
- **Statistical Analysis:** Perform statistical analysis post-loading, calculating mean, standard deviation, and feature distribution. This step uncovers patterns, outliers, and potential anomalies, aiding in tailoring fraud detection models.
- **Feature Scaling:** Apply feature scaling for consistent feature scales during model training. Techniques like normalization ensure similar scales, especially crucial for

TABLE I
CREDIT CARD DATASET DESCRIPTIONS

Attribute	Description
Merchant id	An identifier for the merchant where the transaction took place.
Average Amount/transaction/day	The daily avg transaction amount for that particular merchant.
Transaction amount	The monetary value of the individual transaction.
Is declined	A binary indicator whether the transaction was declined (1) or not (0).
Total Number of declines/day	The total number of transaction declines per day for the merchant.
isForeignTransaction	A binary indicator that represents whether the transaction occurred in a foreign country (1) or not (0).
isHighRiskCountry	A binary indication that indicates whether or not the transaction happened in a nation with a high level of risk (1) or not (0).
Daily chargeback avg amt	The daily avg of chargebacks for that particular merchant.
6 month avg chbk amt	The avg quantity of chargebacks for the merchant over a period of six months.
6-month chbk freq	The frequency of chargebacks for the merchant during a 6 month period.
isFraudulent	A binary label indicating whether the transaction is fraudulent (1) or legitimate (0).

models like Hidden Naive Bayes (HNB) and Bayesian Belief Network (BBN).

- **Normalization:** Transform data to a standardized range (e.g., 0 to 1) through normalization. It's beneficial for models relying on specific feature ranges, enhancing their performance.
- **Random Over Sampling:** Dataset with class imbalance, where the number of normal (non-fraudulent) transactions significantly outweighs fraudulent transactions, can lead to biased models. Random Oversampling was used to address this problem. This involves increasing the number of instances of the minority class (fraudulent transactions) by either duplicating existing data or generating synthetic samples. By balancing the dataset in this manner, performance of the fraud detection models can be improved.
- **Model Training:** The two distinct models for credit card fraud detection: a) Hidden Naive Bayes (HNB) and b) Bayesian Belief Network (BBN) were chosen for their ability to model probabilistic dependencies between variables, which is relevant in capturing the complex patterns associated with credit card fraud. The models are trained using the preprocessed and balanced dataset obtained from the previous steps.
- **Model Testing:** Test the trained models, HNB and BBN, using a separate dataset to assess their real-world effectiveness in detecting credit card fraud.
- **Model Evaluation:** Evaluate HNB and BBN models

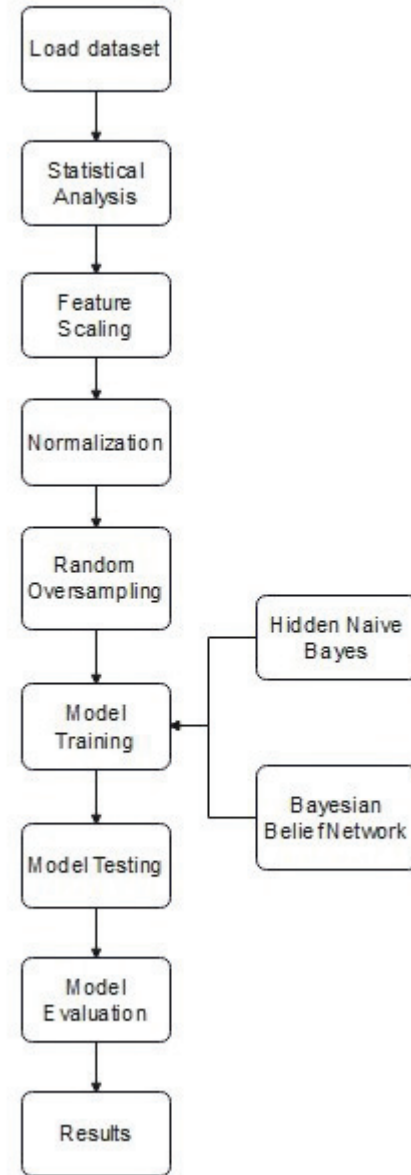


Fig. 1. Block Diagram for Credit Card Fraud Detection

using accuracy, precision, recall, and F1 score metrics to assess their performance in identifying fraudulent transactions. Results guide the models' reliability and effectiveness in fraud detection.

C. Models

- **Hidden Naive Bayes:** HNB extends the Naive Bayes algorithm by incorporating hidden variables into the model to capture dependencies between observed features and class labels. While traditional Naive Bayes assumes feature independence given the class label, Hidden Naive Bayes introduces hidden variables representing unobservable factors that influence observed data. It operates on a generative process where these hidden variables

impact observed features, subsequently affecting class labels. By considering these hidden variables, the model can capture intricate dependencies, especially in cases with unobservable factors impacting observed features, enhancing classification accuracy.

Algorithm:

Input:

D : Training dataset with labeled credit card transactions.

C : Set of possible class labels (fraudulent or non-fraudulent).

Output:

HNB model for credit card fraud detection.

For each class label c in C :

Compute prior probability $P(c)$ based on the frequency of class c in D .

For each pair of attributes A_i and A_j :

Derive conditional probability $P(a_i, a_j | c)$ from D for possible assignments a_i, a_j , and c to A_i, A_j , and C .

For each pair of attributes A_i and A_j :

Calculate interaction parameter $IP(A_i, A_j | C)$ based on conditional probabilities.

For each attribute A_i :

Determine W_i as the sum of interaction parameters over all attributes A_j where $j \neq i$.

For each attribute A_j (where $j \neq i$):

Calculate W_{ij} as the interaction parameter between attributes A_i and A_j .

Output the HNB model including prior probabilities $P(c)$, conditional probabilities $P(a_i, a_j | c)$, interaction parameters $IP(A_i, A_j | C)$, and weight vectors W_i and W_{ij} .

- **Bayesian Belief Network:** A BBN, also known as a probabilistic graphical model, is a graphical representation of probabilistic relationships among a set of variables. It provides a compact and intuitive way to model and reason about uncertainties and dependencies in a domain. A BBN consists of two main components: nodes and edges. Nodes represent variables or events in the domain, and edges represent probabilistic dependencies between the variables. The edges are directed, indicating the direction of influence or dependency. They provide a flexible and intuitive framework for probabilistic modeling and reasoning.

Algorithm:

Input:

- D : A training dataset consisting of labeled credit card transactions.
- C : The set of possible class labels (fraudulent or non-fraudulent).
- A : The set of attributes extracted from credit card transactions.

Output:

- BBN model for credit card fraud detection.
- 1) Initialize an empty Bayesian Belief Network (BBN).
- 2) Create nodes in BBN for each attribute A_i in A .

- 3) Create class nodes in BBN for each class label c in C .
- 4) Connect attribute nodes A_i to class nodes c indicating conditional dependence.
- 5) Analyze training data D to estimate conditional probabilities $P(A_i | A_j, c)$ for each class label c .
- 6) Create arcs connecting attribute nodes A_j to A_i , indicating conditional dependence.
- 7) Determine the appropriate probability distribution for each attribute based on its characteristics.
- 8) Estimate distribution parameters (e.g., mean and variance).
- 9) Initialize evidence variables for observed attributes during testing.

Inference:

- 1) Use probabilistic inference algorithms to propagate beliefs through the network.
- 2) Compute posterior probabilities of class labels based on observed attribute values.

Decision:

- 1) Classify a transaction as fraudulent or non-fraudulent based on posterior probabilities compared against a predefined decision threshold.

Output the BBN model, which includes:

- The network structure (nodes and arcs) representing attribute dependencies.

IV. RESULTS AND DISCUSSION

The model improved dramatically at identifying complex patterns in identifying frauds. We present a detailed evaluation of the performance of our credit card fraud detection models, Hidden Naive Bayes (HNB) and Bayesian Belief Network (BBN). Through a comprehensive analysis of key performance metrics and visualization, we offer insights into the effectiveness of these models in identifying fraudulent transactions. Our results shed light on the capabilities, strengths, and areas for potential improvement of both models, facilitating informed decisions in deploying fraud detection solutions.

A. Confusion Matrix

Confusion Matrix displays the counts or percentage of true positive (TP), true negative (TN), false positive (FP), and false negative (FN) predictions, used to describe the effectiveness of a classification model. It is frequently used to evaluate a model's performance in machine learning and classification tasks.

The Confusion matrix for Hidden Naive Bayes(HNB) is:

TABLE II
CONFUSION MATRIX FOR HNB

	Actual Fraud	Actual Non-Fraud
Predicted Fraud	23	80
Predicted Non-Fraud	1	511

The Confusion matrix for Bayesian Belief Network(BBN) is:

TABLE III
CONFUSION MATRIX FOR HNB

	Actual Fraud	Actual Non-Fraud
Predicted Fraud	24	60
Predicted Non-Fraud	4	528

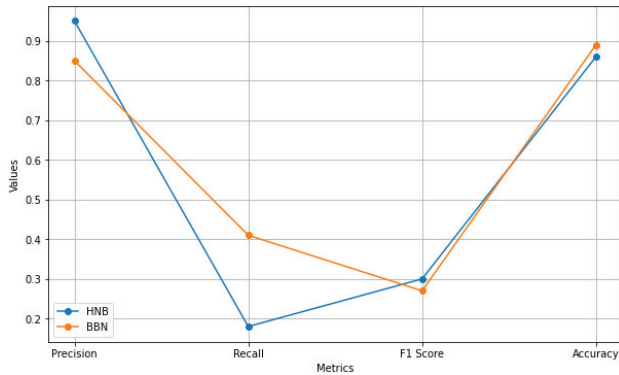


Fig. 2. Comparison of metrics between HNB and BBN models

B. Performance Metrics

When evaluating the model's performance of a credit card fraud detection system, several metrics are commonly used to assess its effectiveness. Here are some important performance metrics:

1) Accuracy:

$$AC = \frac{TN + TP}{TP + TN + FN + FP}$$

2) Precision (PR):

$$PR = \frac{TP}{TP + FP}$$

3) Recall (RC):

$$RC = \frac{TP}{TP + FN}$$

4) F1 Score:

$$F1 \text{ Score} = \frac{2 \cdot PR \cdot RC}{PR + RC}$$

TABLE IV
PERFORMANCE METRICS

	Accuracy	Precision	F1-score	Recall
HNB	86.82	95.8	22.33	0.3622
BBN	89.59	85.18	27.77	0.4181

C. Comparison for Other Models

With differing degrees of accuracy and efficiency, the models that were cited demonstrated encouraging outcomes in the identification of credit card fraud. For example, Xuan et al. [12] demonstrated a Random Forest-based method with excellent results. Using KNN, Khare and Sait [13] produced competitive results. In their investigation of the use of multi-layer perceptrons in fraud transaction detection, Varmedja et al. [14] emphasized the significance of detection accuracy. The study by Ileberi, Sun, and Wang [15] focuses on evaluating how the model's performance can be improved for detection of frauds when ensembling techniques like AdaBoost is used with DT and SMOTE is used to handle class imbalance in the data. An experimental investigation on unbalanced classification techniques for Decision Tree-based credit card fraud detection was carried out in the Makki et al. [16] publication. By contrast, the algorithms of Hidden Naive Bayes (HNB) and Bayesian Belief Network (BBN) shown strong performance in detecting fraudulent transactions. The method made use of the probabilistic modeling capabilities of BBN in conjunction with complex dependency capture using HNB. As a result of this combination, a system that provided scalable and real-time fraud detection was created that could both reliably identify fraudulent activity and effectively handle complicated patterns.

TABLE V
COMPARISON WITH OTHER MODELS

Ref No.	Models	Accuracy
[17]	LR	54.86%
[18]	GA-RF	77.95%
	Proposed Method(HNB)	86.87%
	Proposed Method(BBN)	89.59%

V. CONCLUSION AND FUTURE SCOPE

In this Paper, we addressed the critical issue of credit card fraud detection by leveraging Hidden Naive Bayes (HNB) and Bayesian Belief Network (BBN) models. Both models exhibited commendable accuracy, achieving 86.87% for HNB and 89.59% for BBN, in accurately identifying fraudulent transactions. While these models demonstrated competitive performance compared to alternative methods, our work underscores

the importance of considering diverse evaluation metrics and exploring a range of techniques. Moving forward, our models hold potential for enhancement through feature enrichment, parameter optimization, ensemble techniques, and real-time data stream processing. The future scope also includes the incorporation of additional features like user behavior patterns and device characteristics, the exploration of unsupervised learning for emerging fraud patterns, and the development of transparent and interpretable models. Addressing data imbalance, integrating these models into existing fraud prevention systems, and ensuring continuous monitoring and retraining are essential steps to fortify credit card fraud detection systems against evolving threats. This work contributes to advancing fraud detection, offering robust and accurate solutions for safeguarding financial transactions.

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REFERENCES

- [1] N. K. Trivedi, S. Simaiya, U. K. Lilhore, and S. K. Sharma, "An efficient credit card fraud detection model based on machine learning methods," *Int. J. Adv. Sci. Technol.*, vol. 29, no. 5, pp. 3414–3424, 2020.
- [2] R. Sailusha, V. Gnaneswar, R. Ramesh, and G. R. Rao, "Credit card fraud detection using machine learning," in *Proc. 4th Int. Conf. Intell. Comput. Control Syst. (ICICCS)*, May 2020, pp. 967–972.
- [3] F. Z. El Hloul, J. Riffi, M. A. Mahraz, A. El Yahyaouy, and H. Tairi, "Credit card fraud detection based on multilayer perceptron and extreme learning machine architectures," in *Proc. Int. Conf. Intell. Syst. Comput. Vis. (ISCV)*, Jun. 2020, pp. 1–5.
- [4] K. Randhawa, C. K. Loo, M. Seera, C. P. Lim, and A. K. Nandi, "Credit card fraud detection using AdaBoost and majority voting," *IEEE Access*, vol. 6, pp. 14277–14284, 2018.
- [5] S. Rajora, D. L. Li, C. Jha, N. Bharill, O. P. Patel, S. Joshi, D. Puthal, and M. Prasad, "A comparative study of machine learning techniques for credit card fraud detection based on time variance," in *Proc. IEEE Symp. Comput. Intell. (SSCI)*, Nov. 2018, pp. 1958–1963.
- [6] S. Khatri, A. Arora and A. P. Agrawal, "Supervised Machine Learning Algorithms for Credit Card Fraud Detection: A Comparison," 2020 10th International Conference on Cloud Computing, Data Science & Engineering (Confluence), Noida, India, 2020, pp. 680–683, doi: 10.1109/Confluence47617.2020.9057851.
- [7] V. Joshi, "Abstract data set for credit card fraud detection," 2018. Accessed:(Version 1) April 2020
- [8] International Journal for Research in Applied Science & Engineering Technology (IJRASET) ISSN: 2321-9653; IC Value: 45.98; SJ Impact Factor: 7.538 Volume 10 Issue XI Nov 2022
- [9] Swati Warghade, Shubhada Desai, Vijay Patil, "Credit Card Fraud Detection from Imbalanced Dataset Using Machine Learning Algorithm," *International Journal of Computer Trends and Technology*, vol. 68, no. 3, pp. 22–28, 2020. Crossref, 10.14445/22312803/IJCTT-V68I3P105
- [10] A. Sahu, H. GM and M. K. Gourisaria, "A Dual Approach for Credit Card Fraud Detection using Neural Network and Data Mining Techniques," 2020 IEEE 17th India Council International Conference (INDICON), New Delhi, India, 2020, pp. 1–7, doi: 10.1109/INDICON49873.2020.9342462.
- [11] D. Prajapati, A. Tripathi, J. Mehta, K. Jhaveri and V. Kelkar, "Credit Card Fraud Detection Using Machine Learning," 2021 International Conference on Advances in Computing, Communication, and Control (ICAC3), Mumbai, India, 2021, pp. 1–6, doi: 10.1109/ICAC353642.2021.9697227.
- [12] S. Makki, Z. Assaghir, Y. Taher, R. Haque, M. -S. Hacid and H. Zeineddine, "An Experimental Study With Imbalanced Classification Approaches for Credit Card Fraud Detection," in *IEEE Access*, vol. 7, pp. 93010–93022, 2019, doi: 10.1109/ACCESS.2019.2927266.
- [13] S. Xuan, G. Liu, Z. Li, L. Zheng, S. Wang and C. Jiang, "Random forest for credit card fraud detection," 2018 IEEE 15th International Conference on Networking, Sensing and Control (ICNSC), Zhuhai, China, 2018, pp. 1–6, doi: 10.1109/ICNSC.2018.8361343.
- [14] Khare, N. and Sait, S.Y., 2018. Credit card fraud detection using machine learning models and collating machine learning models. *International Journal of Pure and Applied Mathematics*, 118(20), pp.825–838.
- [15] D. Varmedja, M. Karanovic, S. Sladojevic, M. Arsenovic and A. Anderla, "Credit Card Fraud Detection - Machine Learning methods," 2019 18th International Symposium INFOTEH-JAHORINA (INFOTEH), East Sarajevo, Bosnia and Herzegovina, 2019, pp. 1–5, doi: 10.1109/INFOTEH.2019.8717766.
- [16] E. Ileberi, Y. Sun and Z. Wang, "Performance Evaluation of Machine Learning Methods for Credit Card Fraud Detection Using SMOTE and AdaBoost," in *IEEE Access*, vol. 9, pp. 165286–165294, 2021, doi: 10.1109/ACCESS.2021.3134330.
- [17] J. O. Awoyemi, A. O. Adetunmbi and S. A. Oluwadare, "Credit card fraud detection using machine learning techniques: A comparative analysis," 2017 International Conference on Computing Networking and Informatics (ICCNI), Lagos, Nigeria, 2017, pp. 1–9, doi: 10.1109/ICCNI.2017.8123782.
- [18] Seera M, Lim CP, Kumar A, Dhamotharan L, Tan KH. An intelligent payment card fraud detection system. *Ann Oper Res*. 2021 Jun 8:1–23. doi: 10.1007/s10479-021-04149-2. Epub ahead of print. PMID: 34121790; PMCID: PMC8186361.